

Anish Kanade

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SUMMARY

Full-stack Software Engineer passionate about building **end-to-end web and cloud systems**. Proficient in Java and Python (Django) with strong experience in **front-end development** using JavaScript/TypeScript. Skilled in designing and deploying **AWS-based architectures** (S3, EC2) and delivering **production-ready applications**. Applied **AI/ML** experience developing **CNN and LSTM pipelines** (TensorFlow/Keras), automating dataset labeling with Amazon Mechanical Turk, and extracting facial action-unit features for research-driven insights.

WORK EXPERIENCE

Software Engineer (AI/ML)

April 2024 - Current

University of Massachusetts Boston

- Built and managed a **Django-based** platform that analyzes students' facial expressions to help professors determine whether students have understood the topic, **enhancing classroom insights and data accessibility**.
- Implemented an **end-to-end data pipeline** including preprocessing and a large-scale annotation workflow on **Amazon Mechanical Turk (supporting 500+ facial images)** with built-in quality checks, which **increased labeled coverage** and **improved** the robustness of model training.
- Developed and trained **emotion-recognition models** combining convolutional **neural networks and LSTMs (TensorFlow/Keras)** with attention, producing sequence-level predictions that improved the reliability of emotion detection for research experiments.
- Created a **reusable pipeline** for extracting and storing facial action-unit features, providing consistent, analysis-ready inputs that accelerated experimentation and model iteration.

Orientation Leader

May 2022 - August 2022

University of Massachusetts Boston

- Organized orientation events for 300+ students, ensuring seamless campus integration.
- Mentored 250+ students with individualized academic guidance, improving first-semester retention rates.
- Trained and led a team of 20 orientation assistants, overseeing logistical coordination and ensuring consistent, high-quality program delivery.

PROJECTS

Cloud-Powered Stock Analysis (Python, AWS, Pandas, Matplotlib, Twelve Data API)

- Designed and automated a data pipeline to retrieve, process, and **visualize five years of stock data with 98% accuracy and consistency**.
- Hands-on experience with AWS S3 & EC2 for seamless, cloud integration and real-time visualization updates, **reducing latency by 50%**.
- Utilized the Twelve Data API to ensure reliable financial data acquisition and deliver actionable insights.
- Enhanced expertise in Data analytics and cloud automation, laying the groundwork for predictive modeling to **boost decision-making by 30%**.

Bank Churn Analysis (Python, Pandas, Matplotlib, Scikit-learn)

- Conducted an in-depth customer churn analysis using key customer metrics, identifying attrition factors.
- Built a Logistic Regression model in Scikit-learn that **achieved 81% test accuracy**; validated via 5-fold cross-validation.
- Developed predictive models, **achieving a 15% churn reduction** through actionable stakeholder recommendations.

Recommendation Platform For Hikers: "HikeMates" (React, CSS, JavaScript, HTML, Git, Agile)

- Architected and built responsive web components, **reducing layout-related issues by 35%**.
- Refactored **20+ reusable components**, applying DRY principles and CSS Modules to improve maintainability.
- Led Agile sprint planning** and code review sessions; collaborated with the backend team to integrate RESTful endpoints.
- Collaborated in a **Git-based team environment**, resolving merge conflicts and maintaining a clean version history.

Percolation Simulation: Vacancy vs. Percolation Probability (Java)

- Analyzed complex percolation problems in Java using scientific principles to model propagation phenomena.
- Conducted descriptive Analysis to generate plots that **revealed a 35% correlation between vacancy and probability**.

Markov Text Generator: Probabilistic Sequence Modeling (Python, NumPy)

- Utilized Python with **Object-Oriented programming principles** and the NumPy library for efficient matrix operations.
- Analyzed extensive training data to generate realistic text sequences.
- Demonstrated expertise in Natural Language Processing while **optimizing the model for scalability and high performance with large datasets**.

ACHIEVEMENTS

- ACM SIGGRAPH 2024 – Conference Presentation:** Co-presented "An Iterative Approach to Build a Semantic Dataset for Facial Expression of Personality," showcasing methodology advancements in HCI.
- ACM SIGGRAPH 2024 – Publication:** "An Iterative Approach to Build a Semantic Dataset for Facial Expression of Personality."

EDUCATION

Bachelor of Science in Computer Science

May 2025

University of Massachusetts Boston

SKILLS

Languages and Frameworks: Java, Python, C, C++, SQL, Racket, TypeScript, JavaScript (HTML, CSS, React, Node.js)

Machine Learning: TensorFlow/Keras, Scikit-learn, XGBoost, CNNs, LSTMs, Data Labeling (Amazon Mechanical Turk)

Cloud and DevOps: AWS (S3, EC2, Lambda), Google Cloud Platform(GCP), Microsoft Azure, Docker, RESTful API development

Tools and Methodologies: Git, IntelliJ, Tableau, Agile development, Unity, Amazon Mechanical Turk, Pandas, Matplotlib, Jupyter Notebook, VS Code

Soft Skills: Leadership, Critical Thinking, Analytical Thinking, Project Management, Team Collaboration.